Introduction

🚀 MeterFlow – Usage-Based API Billing Platform

Full Stack Implementation Guide

📌 Project Overview

MeterFlow is a SaaS platform that allows developers to:

- Create APIs
- Generate API keys
- Track usage (per request)
- Apply rate limiting
- Calculate billing based on usage

This simulates real-world systems used by API companies and SaaS platforms.

🧱 Tech Stack

🔵 MERN Stack

- Frontend: React + Tailwind + React Query
- Backend: Node.js + Express
- Database: MongoDB
- Realtime/Queue: Socket.io, BullMQ
- Cache: Redis

🗄️ Database Options (Use-case Based)

1️⃣ MongoDB (NoSQL)

- Best for: Flexible schema, logs, fast iteration
- Use for: API logs, usage tracking

👉 https://www.mongodb.com/

2️⃣ PostgreSQL (Relational)

- Best for: Billing, transactions, structured data
- Strong ACID compliance

👉 https://www.postgresql.org/

3️⃣ MySQL

- Best for: Traditional applications
- Easy hosting + wide support

👉 https://www.mysql.com/

4️⃣ Redis (In-Memory DB)

- Best for:
  - Rate limiting
  - Caching
  - Session storage

👉 https://redis.io/

5️⃣ Elasticsearch

- Best for:
  - Log searching
  - Analytics
  - Fast querying of large datasets

👉 https://www.elastic.co/elasticsearch/

🌐 Free Public APIs for Testing

Use these APIs to simulate usage billing:

1. Pokémon API

https://pokeapi.co/

2. JSON Placeholder (Fake REST API)

https://jsonplaceholder.typicode.com/

3. OpenWeather API

https://openweathermap.org/api

4. CoinGecko Crypto API

https://www.coingecko.com/en/api

5. Dummy Products API

https://dummyjson.com/

🧠 Advanced Features (Optional but Powerful)

- Webhooks for billing alerts
- API analytics (latency, error rate)
- Multi-tenant architecture
- API playground
- Audit logs
- Role-based dashboards

📁 Suggested Folder Structure (MERN)

```
backend/
 ├── controllers/
 ├── routes/
 ├── middleware/
 ├── models/
 ├── services/
 ├── utils/
 ├── config/

frontend/
 ├── components/
 ├── pages/
 ├── hooks/
 ├── services/
 ├── store/
```

💼 Resume Description

Project: MeterFlow – API Billing Platform

- Built a scalable API gateway system to track and meter API usage
- Implemented rate limiting, API key authentication, and request logging
- Designed a billing engine for usage-based pricing
- Developed real-time analytics dashboard using React
- Integrated Stripe for subscription-based payments

🚀 Final Advice

Focus on:

- Gateway logic (this is your USP)
- Clean architecture
- Real-world features (billing, limits, analytics)

Avoid:

- Overcomplicating UI
- Ignoring backend performance

Explanation of Project

🚀 Project Name (Portfolio Ready)

MeterFlow – Usage-Based API Billing & Metering Platform

Inspired by:

- Stripe Billing
- RapidAPI
- AWS API Gateway
- OpenAI

🧠 What You Are Building (Simple Explanation)

A platform where developers can:

- Create APIs
- Give API keys to users
- Track how many requests each user makes
- Charge users based on usage

🏗️ Core System Architecture (High-Level)

🔁 Flow:

1. User signs up
2. Creates an API
3. Gets an unique API key
4. End users hit the API
5. System logs every request
6. Usage is tracked
7. Billing is calculated
8. User pays

⚙️ CORE MODULES (Important)

1️⃣ Authentication & User System

- JWT + Refresh tokens
- Roles:
  - Admin
  - API Owner
  - Consumer

2️⃣ API Management System

- Create API
- Generate API keys
- Revoke keys
- Rotate keys (important 🔥)

3️⃣ API Gateway Layer (🔥 MOST IMPORTANT)

This is your **killer feature**.

Instead of calling APIs directly:

👉 All requests go through YOUR gateway

Responsibilities:

- Validate API key
- Apply rate limiting
- Log request
- Forward request to actual API

4️⃣ Usage Tracking System

Store:

- API key
- Endpoint
- Timestamp
- Response status
- Latency

5️⃣ Billing Engine

Pricing Example:

- Free: 1000 requests
- Pro: ₹0.5 / 100 requests

Logic:

- Count usage per user
- Apply pricing
- Generate invoice

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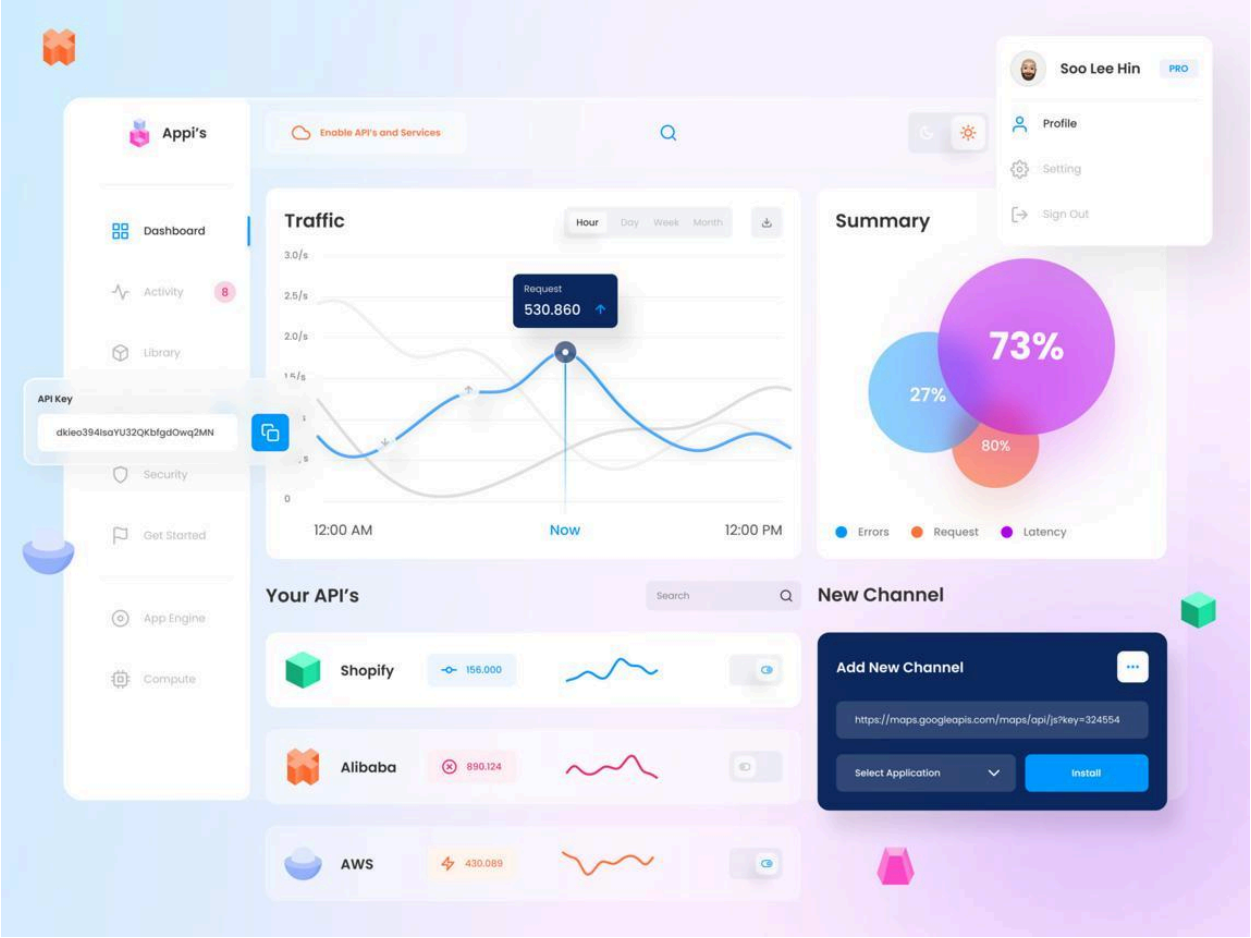
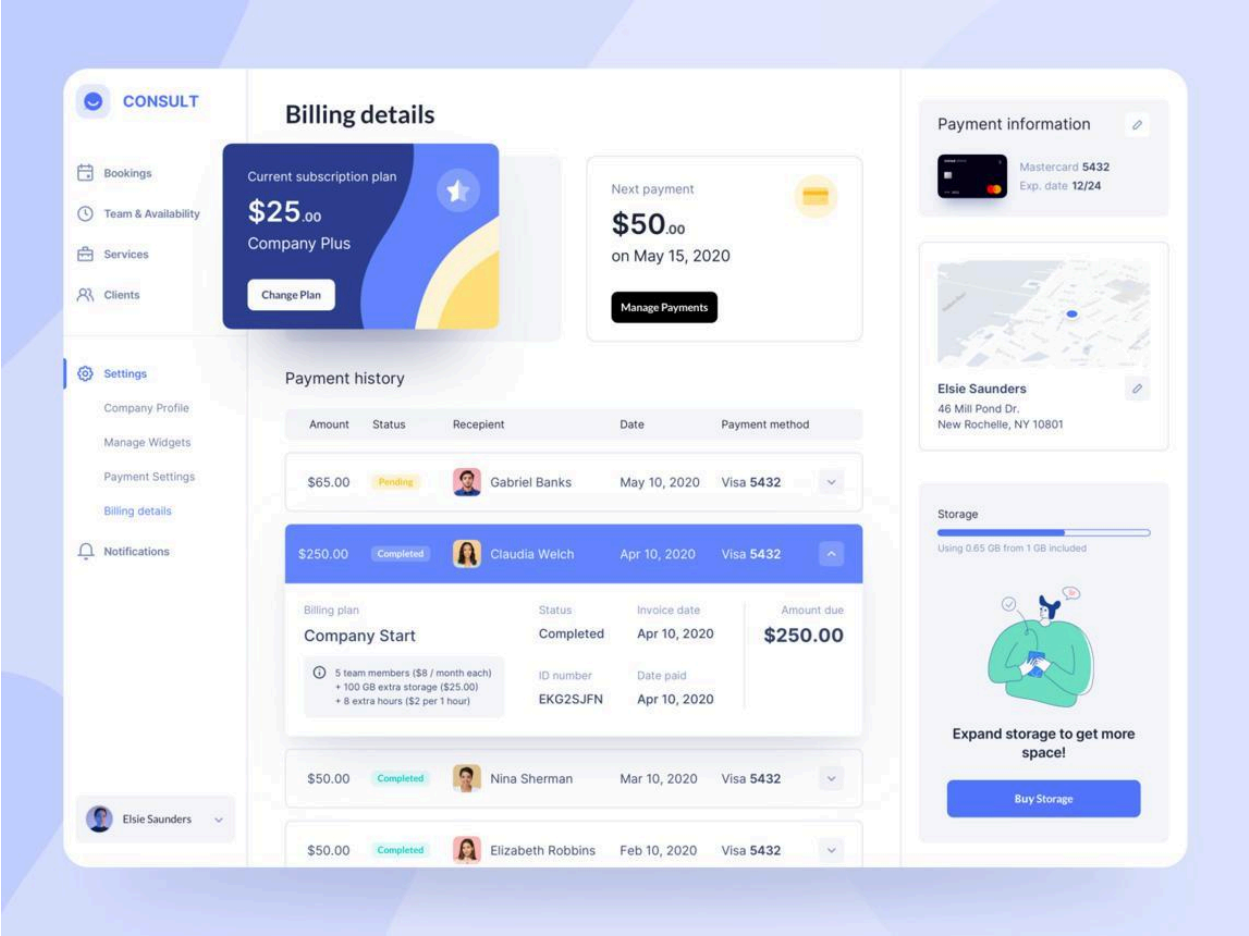
6 Dashboard (Frontend)




CONSULT
Bookings
Team & Availability
Services
Clients
Settings
Company Profile
Manage Widgets
Payment Settings
Billing details
Notifications
Elsie Saunders
Billing details
Current subscription plan
$25.00
Company Plus
Change Plan
Next payment
$50.00
on May 15, 2020
Manage Payments
Payment history
Amount
Status
Recepient
Date
Payment method
$65.00
Pending
Gabriel Banks
May 10, 2020
Visa 5432
$250.00
Completed
Claudia Welch
Apr 10, 2020
Visa 5432
Billing plan
Company Start
5 team members ($8 / month each)
+ 100 GB extra storage ($25.00)
+ 8 extra hours ($2 per 1 hour)
Status
Completed
Invoice date
Apr 10, 2020
Amount due
$250.00
ID number
EKG2SJFN
Date paid
Apr 10, 2020
$50.00
Completed
Nina Sherman
Mar 10, 2020
Visa 5432
$50.00
Completed
Elizabeth Robbins
Feb 10, 2020
Visa 5432
Payment information
Mastercard 5432
Exp. date 12/24
Elsie Saunders
46 Mill Pond Dr.
New Rochelle, NY 10801
Storage
Using 0.65 GB from 1 GB included
Expand storage to get more space!
Buy Storage

7

Show:

- Total requests
- Active API keys
- Usage graph
- Errors
- Revenue

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7 Payment Integration

- Stripe / Razorpay
- Subscription plans
- Payment history

🧱 Tech Stack (Strong Resume)

Frontend

- React
- Tailwind
- Chart.js / Recharts
- React Query

Backend

- Node.js
- Express
- MongoDB (with indexing 🔥)

Advanced

- Redis → Rate limiting
- Nginx → Reverse proxy
- Docker → Deployment
- Queue (BullMQ) → Billing jobs

🧩 Database Design (Important for Interviews)

Users

- id, email, password, role

APIs

- id, userId, name, baseUrl

API Keys

- id, apiId, key, status

Usage Logs

- id, apiKey, endpoint, timestamp, status

Billing

- userId, totalRequests, amount, status

🔥 Advanced Features (This Will Make You Stand Out)

Add these after MVP:

1. Rate Limiting

- Per API key
- Per plan

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2. API Analytics

- Requests per minute
- Error rates
- Latency

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3. Webhooks

- Notify user when limit reached

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4. API Playground

- Test API inside dashboard

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5. Multi-Tenant Architecture

- Each company isolated

💼 Freelancing Angle (VERY IMPORTANT)

You can sell this as:

> "I build API monetization platforms for SaaS companies."

Clients:

- AI startups
- SaaS founders
- Backend-heavy startups

🧠 How to Explain in Interview

Say:

> "I built a usage-based billing system with an API gateway that intercepts and tracks every request, applies rate limiting, and calculates billing dynamically based on usage."

That sounds like a **mid-level backend engineer**.

⚡ Pro Tip (Important)

Most people will:
❌ Just count requests

YOU should:
✅ Build a **gateway layer**

That's what makes this project elite.

📅 2-WEEK EXECUTION PLAN

🗓️ 2-WEEK EXECUTION PLAN

📅 Week 1 – Core Backend + Gateway

Day 1–2: Auth System

- JWT login/signup
- Role system
- API key generation

Day 3–4: API Management

- Create APIs
- Generate/revoke keys
- Store API metadata

Day 5–6: API Gateway (CORE)

- Middleware to:
    - Validate API key
    - Log request
    - Forward request

Day 7: Logging System

- Store:
    - API key
    - Endpoint
    - Timestamp
    - Response status

📅 Week 2 – Billing + Dashboard + Deployment

Day 8–9: Usage Tracking + Billing Engine

- Count requests per user
- Apply pricing logic

Day 10–11: Dashboard (Frontend)

- Usage charts
- API key management
- Billing summary

Day 12: Rate Limiting + Redis

- Limit requests per API key

Day 13: Payment Integration

- Stripe / Razorpay
- Subscription plans

Day 14: Deployment

- Backend → Render / Railway
- Frontend → Vercel
- DB → Mongo Atlas

Resources

🔵 MERN Stack Resources (Node.js + React + MongoDB)

📽️ YouTube (Backend + System Concepts)

- Hitesh Choudhary – Node.js Backend Series
  👉 Covers JWT, APIs, middleware, production practices
- Traversy Media
  👉 “Node.js API Authentication”, “REST API Crash Course”
- Fireship
  👉 Short but powerful videos on system design, scaling, backend concepts

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📦 GitHub Repositories

- **Bulletproof Node.js Architecture**
  https://github.com/santiq/bulletproof-nodejs
  👉 Production-level folder structure (very important)
- **Node.js API Boilerplate**
  https://github.com/hagopj13/node-express-boilerplate
  👉 Auth, validation, structure ready
- **Rate Limiting Example**
  https://github.com/nfriedly/express-rate-limit

📘 Documentation (Must Read)

- Express.js
  👉 https://expressjs.com/
- Mongoose
  👉 https://mongoosejs.com/docs/
- Redis
  👉 https://redis.io/docs/
- Stripe
  👉 https://stripe.com/docs

⚙️ Key Topics to Focus (MERN)

- Express Middleware (for API Gateway 🔥)
- Rate Limiting with Redis
- MongoDB Indexing (for logs)
- Background Jobs (BullMQ)

🌐 API Data Sources (For Testing Your Platform)

Use these APIs to simulate real usage billing:

- PokeAPI
  👉 https://pokeapi.co/
- JSONPlaceholder
  👉 https://jsonplaceholder.typicode.com/
- OpenWeather
  👉 https://openweathermap.org/api
- CoinGecko
  👉 https://www.coingecko.com/en/api
- DummyJSON
  👉 https://dummyjson.com/

Project Submission requirements

github link - backend github link & frontend github link
deploy - backend deploy link & frontend deploy link
demo video - screen record of frontend
documentation - read.md - with ss of frontend